

VecQ: Minimal Loss DNN Model Compression With Vectorized Weight Quantization

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Abstract—Quantization has been proven to be an effective method for reducing the computing and/or storage cost of DNNs. However, the trade-off between the quantization bitwidth and final accuracy is complex and non-convex, which makes it difficult to be optimized directly. Minimizing direct quantization loss (DQL) of the coefficient data is an effective local optimization method, but previous works often neglect the accurate control of the DQL, resulting in a higher loss of the final DNN model accuracy. In this paper, we propose a novel metric, called Vector Loss. Using this new metric, we decompose the minimization of the DQL to two independent optimization processes, which significantly outperform the traditional iterative L2 loss minimization process in terms of effectiveness, quantization loss as well as final DNN accuracy. We also develop a new DNN quantization solution called VecQ, which provides minimal direct quantization loss and achieve higher model accuracy. In order to speed up the proposed quantization process during model training, we accelerate the quantization process with a parameterized probability estimation method and template-based derivation calculation. We evaluate our proposed algorithm on MNIST, CIFAR, ImageNet, IMDB movie review and THUCNews text data sets with numerical DNN models. The results demonstrate that our proposed quantization solution is more accurate and effective than the state-of-the-art approaches yet with more flexible bitwidth support. Moreover, the evaluation of our quantized models on Saliency Object Detection (SOD) tasks maintains comparable feature extraction quality with up to $16\times$ weight size reduction.

Index Terms—DNN compression, DNN quantization, vectorized weight quantization, low bitwidth, vector loss.

1 INTRODUCTION

DEEP Neural Networks (DNNs) have been widely adopted in machine learning based applications [1], [2]. However, besides DNN training, DNN inference is also a computation-intensive task which affects the effectiveness of DNN based solutions [3], [4], [5]. Neural network quantization employs low precision and low bitwidth data instead of high precision data for the model execution. Compared to the DNNs with floating point with 32-bit width (FP32), the quantized model can achieve up to $32\times$ compression rate with an extremely low-bitwidth quantization [6]. The low-bitwidth processing, which reduces the cost of the inference by using less memory and reducing the complexity of the

multiply-accumulate operation, improves the efficiency of the execution of the model significantly [5], [7].

However, lowering the bitwidth of the data often brings accuracy degradation [4], [8], [9]. This requires the quantization solution to balance between computing efficiency and final model accuracy. However, the quantitative trade-off is non-convex and hard to optimize – *the impact of the quantization to the final accuracy of the DNN models is hard to formulate.*

Previous methods neglect the quantitative analysis of the Direct Quantization Loss (DQL) of the weight data and make the quantization decision empirically while directly evaluating the final model accuracy [6], [10], [11], [12], [13] thus only achieving unpredictable accuracy.

In order to achieve higher training accuracy, finding an optimal quantization solution with minimal loss during the training of the learning kernels is effective and practical. One way of finding a local optimal solution is to minimize the DQL of the weight data, which is widely used in the current quantization solutions [14], [15], [16], [17], [18].

As shown in Fig. 1, w_f denotes the full-precision weight and w_q is the value after quantization. Conventional quantization methods regard w_f as a point (set as origin in Fig. 1) in Euclidean Space, and w_q is a point which is close to w_f in a discrete data space. The discrete data space contains a certain number of data points that can be represented by the selected bitwidth. Therefore, the Square of Euclidean Distance (Square 2-norm or called L2 distance [19]), represented as $\|w_f - w_q\|_2^2$, between the original weight data and the quantized data is simply used as the loss of the quantization process, which is going to be reduced [15], [16], [17], [18].

Although the L2 based solutions are proven to be effective

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